

# MATH - 302

## Chapter 23 to 45

### Formulae

1) Average = 10, 7, 9, 27, 2 → Find Average  
option (a) :  $AVERAGE(A4:A8) = 11$

OR

$$OPTION(B) = \frac{10 + 7 + 9 + 27 + 2}{5} = 10.6 = 11 \text{ (Round off)}$$

2) Median = 1, 2, 3, 4, 5, 6 → Find median

$$(a) = \text{Median}(A14:A19) \\ = 3.5 \text{ (middle value)}$$

OR

$$(b) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

3)

3) Mode -

Most repeated value of the data is called Mode.

1, 2, 3, 3, 4, 5, 6.

3 Mode

For Excel =  $MODE(A27:A32)$

1) Frequency - (data - array, bins - array)

score

Bins

79

70

85

79

78

89

85

50

81

95

88

97

Step 1:- After copying data to a blank excel worksheet, select range A1:B16 starting with formula cell.

Step 2:- Press F2

Step 3:- Press CTRL + SHIFT + ENTER

① = FREQUENCY (A2:A10, B2, B5) → simultaneously enter  
↓  
CTRL + SHIFT + ENTER

②

④

①

data.

less than or equal to 70

bins 71 to 79

bins 80 to 89

Greater than or equal to 90

### 5) Count Function :-

For Excel.

$$= \text{Count}(\text{value 1, value 2, } \dots)$$

### 6) Arithmetic Mean for Grouped Data

$$A.M = \frac{\sum fX}{n}$$

### 7) Geometric Mean = $G.M = (x_1 \times x_2 \times x_3)^{1/n}$

$$G.M = (x_1 \times x_2 \times x_3)^{1/n}$$

### 8) Harmonic Mean

$$H.M = n / (1/x_1 + 1/x_2 + 1/x_3 \dots)$$

Note =  $n \rightarrow$  mean total # of value in data.

9) Quartiles (It divides into 4 equal parts)  
For symmetrical or ungrouped data

$$1^{st} \text{ Quartile} = Q_1 = \frac{(n+1)}{4}$$

$$2^{nd} \text{ Quartile} = Q_2 = \frac{2(n+1)}{4}$$

$$3^{rd} \text{ Quartile} = Q_3 = \frac{3(n+1)}{4}$$

For Grouped Data

~~For Deciles~~

$$Q_i = i^{th} \text{ Quartile} = l + h/f [\text{sum } f/4 \times i - cf]$$

$l$  = low boundary

$h$  = width of CI

$cf$  = Cumulative frequency

10) Deciles for symmetrical or ungrouped data

(It is divided into 10 equal parts)

$$1^{st} \text{ Decile } D_1 = (n+1)/10$$

$$2^{nd} \text{ " } D_2 = 2(n+1)/10$$

$$9^{th} \text{ " } D_9 = 9(n+1)/10$$

For Grouped Data

$$Q_i = i^{th} \text{ Decile } (i=1,2,9) = l + h/f [\text{sum } f/10 \times i - cf]$$

11) Percentile (It divides into 100 equal parts)

For symmetrical or ungrouped data

$$1^{st} \text{ Percentile} = P_1 = (n+1)/100$$

$$2^{nd} \text{ " } = P_2 = 2(n+1)/100$$

$$99^{th} \text{ " } = P_{99} = 99(n+1)/100$$

For Grouped data

$$Q_i = i^{th} \text{ percentile} = l + h/f [\text{sum } f/100 \times i - cf]$$

12) Empirical Relationships:-

~~System~~ \*) symmetrical Distribution

$$\text{Mean} = \text{Median} = \text{Mode}$$

\*) Positive skewed Distribution

$$\text{mean} > \text{median} > \text{mode}$$

\*) Negative skewed Distribution

$$\text{mode} > \text{median} > \text{mean}$$

\*) Moderately skewed and unimodal Distribution

$$\text{mean} - \text{mode} = 3(\text{mean} - \text{median})$$

13) The Mean (Arithmetic Average)

Simple mean

$$\bar{x} = \sum x/n = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Population mean

$$\mu = \sum x/N = \frac{x_1 + x_2 + \dots + x_n}{N}$$

14) Midhinger

$$\text{Midhinger} = \frac{Q_1 + Q_3}{2}$$

15) Quartile Deviation:-

$$Q.D. = \frac{(Q_3 - Q_1)}{2}$$

16) variance / variation:- / Coefficient of Variation

$$C.V = \left( \frac{s}{\bar{x}} \right) 100\%$$

17) Interquartile Range:-

Difference b/w 3rd & 1st

$$Q_3 - Q_1$$

## 18) Variance

For Population:  $\sigma^2 = \frac{\sum (X_i - \mu)^2}{N}$

For the sample:  $s^2 = \frac{\sum (X - \bar{X})^2}{n-1}$

## 19) Standard Deviation

For the population:  $\sigma = \sqrt{\frac{\sum (X - \mu)^2}{N}}$

For the sample:  $s = \sqrt{\frac{\sum (X - \bar{X})^2}{n-1}}$

## 20) Mean Deviation

mean deviation about Mean - Normal Data

$$MD(\text{mean}) = \frac{\sum (x_i - \text{mean})}{n}$$

For Grouped Data

$$MD(\text{mean}) = \frac{\sum f_i (x_i - \text{mean})}{\sum f_i}$$

Mean deviation about median - Normal Data

$$MD(\text{median}) = \frac{\sum (x_i - \text{median})}{n}$$

Grouped Data

$$MD(\text{median}) = \frac{\sum f_i (x_i - \text{median})}{\sum f_i}$$

21) Correlation r

| X  | Y   | $x^2$            | $y^2$ | XY   | $x_m$ | $y_m$ | $\text{Corr}(x,y)$ |
|----|-----|------------------|-------|------|-------|-------|--------------------|
| 2  | 60  | 4                | 3600  | 120  |       |       |                    |
| 5  | 100 | <del>25</del> 25 | 10000 | 500  |       |       |                    |
| 4  | 70  | 16               | 4900  | 280  |       |       |                    |
| 6  | 90  | 36               | 8100  | 540  |       |       |                    |
| 3  | 80  | 9                | 6400  | 240  |       |       |                    |
| 20 | 400 | 90               | 33000 | 1680 | 4     | 80    |                    |

$$SD. = 1.4$$

$$r = 0.8$$

for Excel.

CORREL(array1, array2)

| X         | Y          |
|-----------|------------|
| 2         | 60         |
| 5         | 100        |
| 4         | 70         |
| 6         | 90         |
| 3         | 80         |
| <u>20</u> | <u>400</u> |

$$r = 0.8$$

22) SLOPE (known y's, known x's)

known Y

known X

2

6

3

5

9

11

1

7

8

5

7

4

5

4

0.305556



slope of the line

23) INTERCEPT (known y's, known x's)

known Y

known X

2

6

3

5

9

11

1

7

8

5

0.0483871

24) PERMUT (number, number-chosen)

| <u>Data</u>      | <u>Description</u>           |
|------------------|------------------------------|
| 100              | # of obj.                    |
| 3                | # of obj in each permutation |
| <u>970200</u>    |                              |
| ↓                |                              |
| = PERMUT(A4; A5) |                              |

25) Combinations.

$$nCr = n! / r! (n-r)!$$

26) Binomial Expansion

$$(x+y)^1 = x + y$$

$$(x+y)^2 = x^2 + 2xy + y^2$$

$$(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$(x+y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$$

27) Coefficients of Binomial Expansion:-

$$(x+y)^5 = {}^5C_0 x^5 + {}^5C_1 x^4 y + {}^5C_2 x^3 y^2 + {}^5C_3 x^2 y^3 + {}^5C_4 x y^4 + {}^5C_5 y^5$$

28) Probability :- (Answer should be in %)   
 Percentage

$$P(A \text{ or } B) = P(A) + P(B)$$

$$P(A) + P(B) + P(C) + \dots + P(N) = 1$$

29) ~~BINOMDIST~~ (number-s, trials,

29) BINOMDIST (number-s, trials, probability-s, cumulative)

Data

Description

6

# of successes of trials

10

" " independent trials

0.5

Probability of success on each trial

0.2050781  $\rightarrow$  being successful

30) NEGBINOMDIST (number-f, number-s, probability-s)

31) CRITBINOM (trials, probability-s, alpha)

32) POISSON (x, mean, cumulative)

$$\text{Poisson} = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$33) \text{ CUMPOISSON} = \sum_{k=0}^x \frac{e^{-\lambda} \lambda^k}{k!}$$

34) Calculating Z-values

$$Z = (\text{value} - \text{mean}) / \text{st. Dev.}$$

35) CHITEST (A3 : B5, A7 : B9)

36) Lec → 41 to 48



Probability related.

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$